Class XI
Mathematics (Code- 041)

Time: 3 Hours **MM-80**

Instructions: -

1. This Question paper contains - five sections A, B, C, D and E. Each section is compulsory. However, there are internal choices in some questions.
2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.
3. Section B has 5 Very Short Answer (VSA)-type questions of 2 marks each.
4. Section C has 6 Short Answer (SA)-type questions of 3 marks each.
5. Section D has 4 Long Answer (LA)-type questions of 5 marks each.
6. Section E has 3 source based/case based/passage based/integrated units of assessment (4 marks each) with sub parts.

SECTION A
(Multiple Choice Questions)
Each question carries 1 mark

1. Which of the following is a singleton set?

a) $\{x: |x| < 4, x \in N\}$ b) $\{x: |x| < -4, x \in N\}$

c) $\{x: x^2 = 4, x \in N\}$ d) $\{x: x^2 = 4, x \in R\}$ **1**

2. If $A = \{1,3,5,7,9\}$, then number of proper subsets of A is **1**

a) 31 b) 25 c) 120 d) *none of these*

3. The n^{th} term of a GP 5, 25, 125, ... is **1**

a) 5^n b) 5^{n-1} c) 5^{n+1} d) 5^{n-2}

4. 8^{th}term from the end of the sequence 3, 6, 12, …… 25^{th}term is **1**

a) 393216 b) 393206 c) 313216 d) 303216

5. If $tan\ A = \frac{1}{2}$ and $tan\ B = \frac{1}{3}$, then the value of $A + B$ is

a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) π d) 0 **1**

6. The value of cos 5π is **1**

a) 0 b) 1 c) -1 d) *none of these*

7. 40° 20′ into radian is

a) $\frac{121\pi}{540}$ b) $\frac{120\pi}{540}$ c) $\frac{221\pi}{540}$ d) $\frac{41\pi}{540}$ **1**

8. In an examination there are three multiple choice questions and each question has 4 choices. Number of ways in which a student can fail to get all answer correct is **1**

a) 11 b) 12 c) 27 d) 63

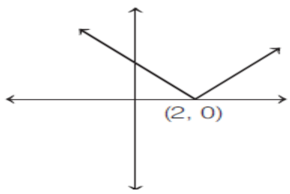
9. If$C(n, 8) = C(n, 2)$ then the value of n **1**

a) 10 b) 8 c) 2 d) 54

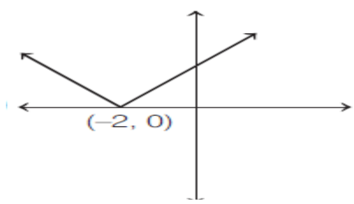
10. The values of a and b, if ordered pair is $(2a - 5, 4) = (5, b + 6)$. **1**

a) $-2,5$ b) $2,5$ c) $5,2$ d) $5, -2$

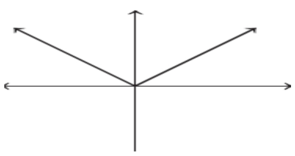
11. The graph of the functions,$f\ (x\ ) = |x - 2|$ is



a)



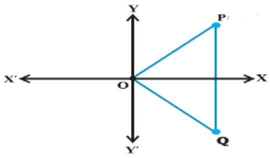
b) **1**



c)

d) *none of these*

12. If P represent complex number $x + iy$ in an Argand plane, then Q represent **1**



a) $-x - iy$ b) $-x + iy$

c) $x - iy$ d) $x + iy$

13. $1 + i^{10} + i^{20} + i^{30}$is a

a) Real Number b) Complex Number **1**

c) Natural Number d) None of these

14. If $x < 5$, then **1**

a) $-x < -5$ b) $-x \le -5$ c) $-x > -5$ d) $-x \ge -5$

15. If $-3x + 17 < -13$, then **1**

a) $x \in (10, \infty)$ b) $x \in [10, \infty)$ c) $x \in (-\infty, 10]$ d) $x \in [-10, 10)$

16. Number of terms in the expansion of $(x + 1)^7$ is **1**

a) 4 b) 8 c) 6 d) 7

17. The fourth term in the expansion of $(x - 2y)^{12}$ is: **1**

a) $-1670x^9 \times y^3$ b) $-1760x^9 \times y^3$ c) $-7160x^9 \times y^3$ d) $-1607x^9 \times y^3$

18. Expansion of $(1 + a)^4 - (1 - a)^4$ is

a) $8a(1 - a^2)$ b) $8a(a^2 - 1)$ c) $8a^2(1 + a)$ d) $8a(1 + a^2)$ **1**

ASSERTION-REASON BASED QUESTIONS

In the following questions, a statement of assertion (A) is followed by a statement of Reason (R). Choose the correct answer out of the following choices.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

19. If set A = {1, 2, 3,4} and B = {3,4, 5,6,7} **1**

Assertion: A ∪ B = {1, 2, 3,4,5,6,7}

Reason: A ∪ B = $\{x\colon x$ belongs to A or B$\}$

20. **Assertion (A):** If the numbers $\frac{-2}{7}, k, \frac{-7}{2}$ are in GP, then $k = \pm 1$. **1**

Reason (R): If $a_1, a_2, a_3$are in GP, then $\frac{a_2}{a_1} = \frac{a_3}{a_2}$.

SECTION B

This section comprises of very short answer type-questions (VSA) of 2 marks each

21. Draw appropriate Venn diagram for each of the following: **2**

$(i)(A \cup B)'$ $(ii)\ (A \cap B)'$

22. Find the sum of the sequence 7, 77, 777, 7777, ... to *n* terms.

OR **2**

If A.M. and G.M. of two positive numbers *a* and *b* are 10 and 8, respectively, find the numbers.

23. In a circle of diameter 40 cm, the length of a chord is 20 cm. Find the length of minor arc of the chord. **2**

24. Solve the inequality and show the graph of the solution on number line.

$$3(1 - x) < 2(x + 4)$$ **2**

OR

Solve the inequality: $\frac{3(x-2)}{5} \le \frac{5(2-x)}{3}$.

25. The longest side of a triangle is 3 times the shortest side and the third side is 2 cm shorter than the longest side. If the perimeter of the triangle is at least 61 cm, find the minimum length of the shortest side. **2**

SECTION C

(This section comprises of short answer type questions (SA) of 3 marks each)

26. Draw the graph of $f(x) = \cos x$, also write Domain and range of the $f(x)$. **3**

27. In how many of the distinct permutations of the letters in MISSISSIPPI do the four I's not come together?

OR

In how many ways can the letters of the word PERMUTATIONS be arranged if the **3**

(i) words start with P and end with S,

(ii) vowels are all together,

(iii) there are always 4 letters between P and S?

28. A committee of 3 persons is to be constituted from a group of 2 men and 3 women. In how many ways can this be done? How many of these committees would consist of 1 man and 2 women? **3**

29. Define greatest integer function. Draw its graph. Also write Domain and Range of the greatest integer function.

OR **3**

Find the domain of the following real functions:

$$\boldsymbol{f(x) = \sqrt{9 - x^2}}$$

30. Find multiplicative inverse of $\sqrt{5} + 3i$.Express the result in the form of $a + ib$. **3**

OR

Express $(1 - i)^4$in the form $a + ib$.

31. Using binomial theorem, evaluate $(99)^5$. **3**

SECTION D

(This section comprises of long answer-type questions (LA) of 5 marks each)

32. *Prove that:*

$$cos\,2x\ \ cos\frac{x}{2} - cos\,3x\ cos\frac{9x}{2} = sin\,5x\ sin\frac{5x}{2}$$

OR **5**

Prove that:

$$\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

33. If $\cos x = \frac{-1}{3}$ $, x\ lies\ in\ quadrant\ III$. Find $sin\ \frac{x}{2}, cos\ \frac{x}{2} and\ tan\ \frac{x}{2}$. **5**

34. If $(x + iy)^3 = u + iv$, then show that $\frac{u}{x} + \frac{v}{y} = 4(x^2 - y^2)$. **5**

35. Show that $9^{n+1} - 8n - 9$ is divisible by 64, whenever *n* is a positive integer.

OR **5**

If *a* and *b* are distinct integers, prove that *a* – *b* is a factor of$a^n - b^n$, whenever *n* is a positive integer.

SECTION E

(This section comprises of 3 case-study/passage-based questions of 4 marks each with two sub-parts. First two case study questions have three sub parts (i), (ii), (iii) of marks 1, 1, 2 respectively. The third case study question has four sub parts 01 marks each.)

36. Read the Case study given below and answer the questions. A state cricket authority has to choose a team of 11 members, to do it so the authority asks 2 coaches of a government academy to select the team members that have experience as well as the best performers in last 15 matches. They can make up a team of 11 cricketers amongst 15 possible candidates

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4

i. In how many ways can the final eleven be selected from 15 cricket players if there is no restriction.

ii. In how many ways can the final eleven be selected from 15 cricket players if two of them being leg spinners, one and only one leg spinner must be included?

iii. If there are 6 bowlers, 3 wicket-keepers, and 11 batsmen in all. The number of ways in which a team of 4 bowlers, 2 wicket-keepers, and 5 batsmen can be chosen.

37. Read the following passage and answer the questions:

Let f and g be two real function defined by f(x) = $\sqrt{x-1}$ and g(x) = $3-2x$ **Based on the above information answer the following questions.** **4**

(i) . Domain of f is

A . $(1,\infty)$ B. $[1,\infty)$ C. $(-\infty,1)$ D. $(-\infty,1]$

(ii) . Domain of 1/g is

A . R $-\left\{\frac{3}{2}\right\}$ B. R C. R $-\left\{\frac{2}{3}\right\}$ D. R $-\left\{-\frac{3}{2}\right\}$

(iii). Domain of f + g

A .R B . R $-(1,\infty)$ C. R $-[1,\infty)$ D. R $-\left\{\frac{2}{3}\right\}$

(iv). Domain of $\frac{g}{f}$ is

A. R $-\left\{\frac{3}{2}\right\}$ B. R C. R - $\left\{\frac{2}{3}\right\}$ D.none of these.

38. Read the following passage and answer the questions:

A mathematics teacher Suman writes the following non-empty sets A and B on a black board while teaching Mathematics in her class. **4**

$A = \{x: x\ is\ a\ letter\ in\ I\ LOVE\ MATHEMATICS\}$

$B = \{x: x\ is\ a\ letter\ in\ I\ LOVE\ STATISTICS\}$

Based on the above information, answer the following questions:

i. Which of the following is true?

a) $A = B$ b) $A \subset B$ c) $B \subset A$ d) $none\ of\ these$

ii. $A \cup B$ is equal to

a) A b) B c) $A \cap B$ d) $\emptyset$

iii. $A \cap B\ is\ equal\ to$

a) A b) B c) $A \cup B$ d) $\emptyset$

iv. Number of subsets of set B is

a) 512 b) 511 c) 1024 d) 1023